

# Relation Algebra

## §10. Freyd’s Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows,  $\exists$  **epsilon**, subject to

$\exists_R \square = R \square, \quad \exists_R = \exists_R \square$   
 $\mathbb{1} \sqsubseteq (R/\exists_R)(\exists_R/R)$   
 $\frac{\exists_R}{\exists_R} = \mathbb{1}$

$\exists_R$  is **thick**  
 $\exists_R$  is **straight**

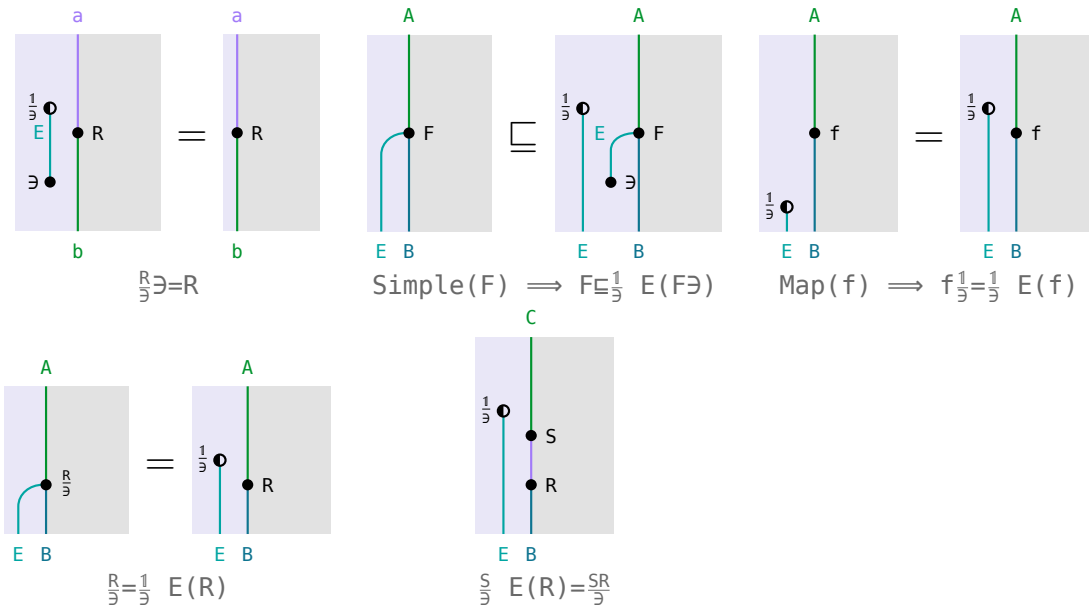
$R \square$  is  $R$ ’s target, an identity arrow. For  $R : A \rightarrow B$  write  $\exists : EB \rightarrow B$ , dropping the subscript.

$\in \triangle \exists^\circ : A \rightarrow EA$

(10b)

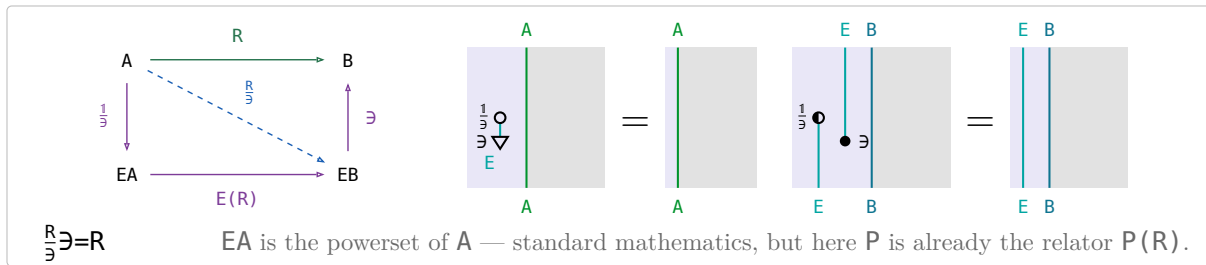
| law                                                                                                                                                                        | the reading                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\exists_R \square = R \square, \quad \exists_R = \exists_R \square$                                                                                                     | One $\exists$ per object, not per arrow.                                                                                                                                                      |
| 2 $\exists$ is <b>thick</b>                                                                                                                                                | <b>Comprehension:</b> every $x$ has a set of exactly the people $x$ admires.                                                                                                                  |
| 3 $\frac{R}{\exists} : A \rightarrow EB$ , for $R : A \rightarrow B$                                                                                                       | convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$                                                                                                                    |
| 4 $\text{Map}(\frac{R}{\exists})$                                                                                                                                          |                                                                                                                                                                                               |
| 5 $\frac{R}{\exists} \exists = R$                                                                                                                                          | reading the chosen set back through $\exists$ returns the relation.                                                                                                                           |
| 6 $\text{Simple}(F) \implies F \sqsubseteq \frac{F \exists}{\exists}$                                                                                                      | A partial choice of sets is inside the total one.                                                                                                                                             |
| 7 the <b>singleton map</b> is monic: $\frac{1}{\exists}$<br>$(\frac{1}{\exists})^\circ \sqsubseteq \mathbb{1}$ two points with the same one-person set are the same point. | The one-person set.                                                                                                                                                                           |
| 8 $\frac{\exists}{\exists} = \mathbb{1}$                                                                                                                                   | Make the set of a set, then read it back one level down.                                                                                                                                      |
| 9 <b>fusion:</b> $\text{Map}(f) \implies \frac{fR}{\exists} = f \frac{R}{\exists}$                                                                                         | Naturality of the unit, $\text{Map}(f) \implies f \frac{1}{\exists} = \frac{1}{\exists} E(f)$ . renaming a point and then taking its one-person set is taking the set and renaming inside it. |
| 10 $\text{Map}(f) \wedge \mathbb{1} \square = \exists \square \implies \frac{f}{\exists} = f \frac{1}{\exists}$                                                            | Rename first or take singletons first — the fusion row above at $R=1$ .                                                                                                                       |
| 11 $R \square = \exists \square \implies R/_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$                                                                             | $x$ and $y$ match when $R$ sends $x$ and $S$ sends $y$ to the same set.                                                                                                                       |
| 12 $E(R) \triangleq \frac{\exists R}{\exists}$                                                                                                                             | $E(R) : EA \rightarrow EB$ , image of a set of $A$                                                                                                                                            |
| 13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$                                                                                                                            | $\frac{1}{\exists} : x \mapsto \{x\}$                                                                                                                                                         |
| 14 $\frac{S}{\exists} E(R) = \frac{SR}{\exists}$                                                                                                                           | absorption — the monad’s composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$ , §Section 10.3                                                                 |
| 15 $\supseteq \triangleq \exists / \exists$                                                                                                                                | $xs \supseteq ys \iff \forall a. \quad ys \exists a \rightarrow xs \exists a$                                                                                                                 |
| 16 $\sqsubseteq = \supseteq^\circ$                                                                                                                                         | $xs \sqsubseteq ys \iff ys \supseteq xs$                                                                                                                                                      |

(10c)



### §10.1. $\mathbf{i} \dashv \mathbf{E}$ Power Allegory defined as adjunction

(10.1a)



$\mathbf{i}$  is the inclusion  $\mathbf{Map}(\mathcal{A}) \rightarrow \mathcal{A}$ , doing nothing, so  $\mathbf{E}$  is both the functor and the monad  $\mathbf{i} \mathbf{E}$ .

### §10.2. $\mathbf{E} \setminus$ as a composite

$$\mathbf{E} \setminus \mathbf{Z} = \subseteq \left( \frac{\mathbf{Z}^\circ}{\mathbf{Z}} \right)^\circ \quad (10.2a)$$

$\mathbf{x} \mathbf{s}$  is related to  $\mathbf{c}$  by  $\mathbf{E} \setminus \mathbf{Z}$  exactly when  $\mathbf{x} \mathbf{s} \subseteq \Lambda(\mathbf{Z}^\circ)(\mathbf{c})$ , the set of every  $\mathbf{a}$  with  $\mathbf{a} \mathbf{Z} \mathbf{c}$

(a)

(b)

(c)

|     |                                                                                            |                                                                                                                                |
|-----|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\mathbf{E} \setminus \mathbf{Z}$                                                          | $\mathbf{Z} = \mathbf{E} \Lambda(\mathbf{Z}^\circ)^\circ$ — (10b)                                                              |
| (b) | $\mathbf{E} \setminus \mathbf{E} \left( \frac{\mathbf{Z}^\circ}{\mathbf{Z}} \right)^\circ$ | $\mathbf{X} \setminus (\mathbf{Y} \mathbf{f}^\circ) = (\mathbf{X} \setminus \mathbf{Y}) \mathbf{f}^\circ$ , $\mathbf{f}$ a map |
| (c) | $\subseteq \left( \frac{\mathbf{Z}^\circ}{\mathbf{Z}} \right)^\circ$                       |                                                                                                                                |

$$\mathbf{R} \setminus \mathbf{S} = \frac{\mathbf{R}^\circ}{\mathbf{S}} \subseteq \left( \frac{\mathbf{S}^\circ}{\mathbf{S}} \right)^\circ \quad (10.2b)$$

$\mathbf{b}$  is related to  $\mathbf{c}$  by  $\mathbf{R} \setminus \mathbf{S}$  exactly when  $\Lambda(\mathbf{R}^\circ)(\mathbf{b}) \subseteq \Lambda(\mathbf{S}^\circ)(\mathbf{c})$ , the  $\mathbf{R}$ -preimage of  $\mathbf{b}$  inside the  $\mathbf{S}$ -preimage of  $\mathbf{c}$

(a)

(b)

(c)

(d)

|     |                                                                                                          |                                                                                                                                                           |
|-----|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\mathbf{R} \setminus \mathbf{S}$                                                                        | $\mathbf{R} = \mathbf{E} \Lambda(\mathbf{R}^\circ)^\circ$ — (10b)                                                                                         |
| (b) | $\mathbf{E} \left( \frac{\mathbf{R}^\circ}{\mathbf{S}} \right)^\circ \setminus \mathbf{S}$               | $(\mathbf{X} \mathbf{Y}) \setminus \mathbf{S} = \mathbf{Y} \setminus (\mathbf{X} \setminus \mathbf{S})$                                                   |
| (c) | $\left( \frac{\mathbf{R}^\circ}{\mathbf{S}} \right)^\circ \setminus (\mathbf{E} \setminus \mathbf{S})$   | $\mathbf{f}^\circ \setminus \mathbf{X} = \mathbf{f} \mathbf{X}$ ; $\mathbf{E} \setminus \mathbf{S} = \subseteq \Lambda(\mathbf{S}^\circ)^\circ$ — (10.2a) |
| (d) | $\frac{\mathbf{R}^\circ}{\mathbf{S}} \subseteq \left( \frac{\mathbf{S}^\circ}{\mathbf{S}} \right)^\circ$ |                                                                                                                                                           |

### §10.3. $\mathcal{A} \cong \mathbf{Kleisli}(E)$

(10.3a)

$E(R) \triangleq \frac{\exists R}{\exists}$ ,  $\frac{1}{\exists} : A \rightarrow EA$ ,  $\text{union} = E(\exists) : E(EA) \rightarrow EA$

$f \diamond g \triangleq f E(g) \text{union}$ , for  $f : A \rightarrow EB$  and  $g : B \rightarrow EC$

the monad is on  $\mathbf{Map}(\mathcal{A})$ , not on the allegory:  $E$  is a relator on all relations, but  $\frac{1}{\exists}$ ,  $\text{union}$  and  $f E(g) \text{union}$  are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{array}{c}
 \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} \xrightarrow{\text{definition of } \diamond, \text{union} = E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y) = E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\exists\right)} \\
 \xrightarrow{(10b), \frac{R}{\exists}\exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{(10b), \text{absorption}} \boxed{\frac{SR}{\exists}}
 \end{array}$$

the first three steps only unfold  $\diamond$  —  $E$  is a functor, so  $E\left(\frac{R}{\exists}\right)\text{union} = E\left(\frac{R}{\exists}\exists\right) = E(R)$  and the  $\text{union}$  is gone.

Absorption, the row of (10b), is the whole law, and it is the functoriality of  $\frac{1}{\exists}$ .

$1$  goes to  $\frac{1}{\exists}$ , the Kleisli identity, by definition — so  $\frac{1}{\exists}$  is an isomorphism of categories  $\mathcal{A} \cong \mathbf{Kleisli}(E)$ , and §Section 10.1's  $i \dashv E$  is the Kleisli adjunction.  $i$  is the identity on objects, so every object of the allegory is free.